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GOSIM Shenzhen 2026

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Building Hybrid AI Systems: Edge Inferencing Meets Cloud Reasoning

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The warehouse floor without AI



SafetyLens v2 Warehouse Safety Visual Alerts Agent – powered by Qwen3-VL deployed on Amazon EKS Hybrid Nodes (NVIDIA DGX Spark)

No feed

Upload Video

/tmp/video.mp4 or rtsp://...

▶ Load

■ Stop

Camera #1

Click to upload

No feed

Camera #2

Click to upload

No feed

Camera #3

Click to upload

No feed

Alert Configuration

Score threshold ≤ 40

Enable alerts:

Webhook URL: https://... (optional)

Safety Query

Ask about cameras or video...

Ask

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01

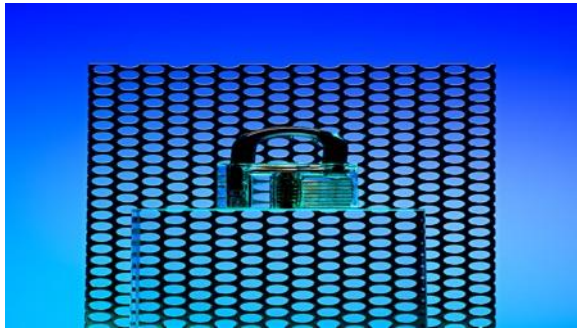
Why Hybrid AI



AI Infrastructure Trends



AI Infrastructure Is Becoming a Strategic Constraint



Sovereign and Regulatory Requirements Shape Deployment Models

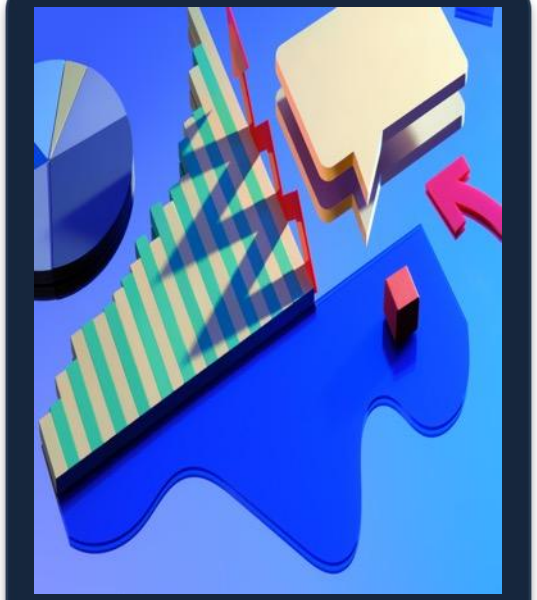


Applications where millisecond latency is nonnegotiable

Integrating AI workloads across cloud, on-premises and edge



Orchestration

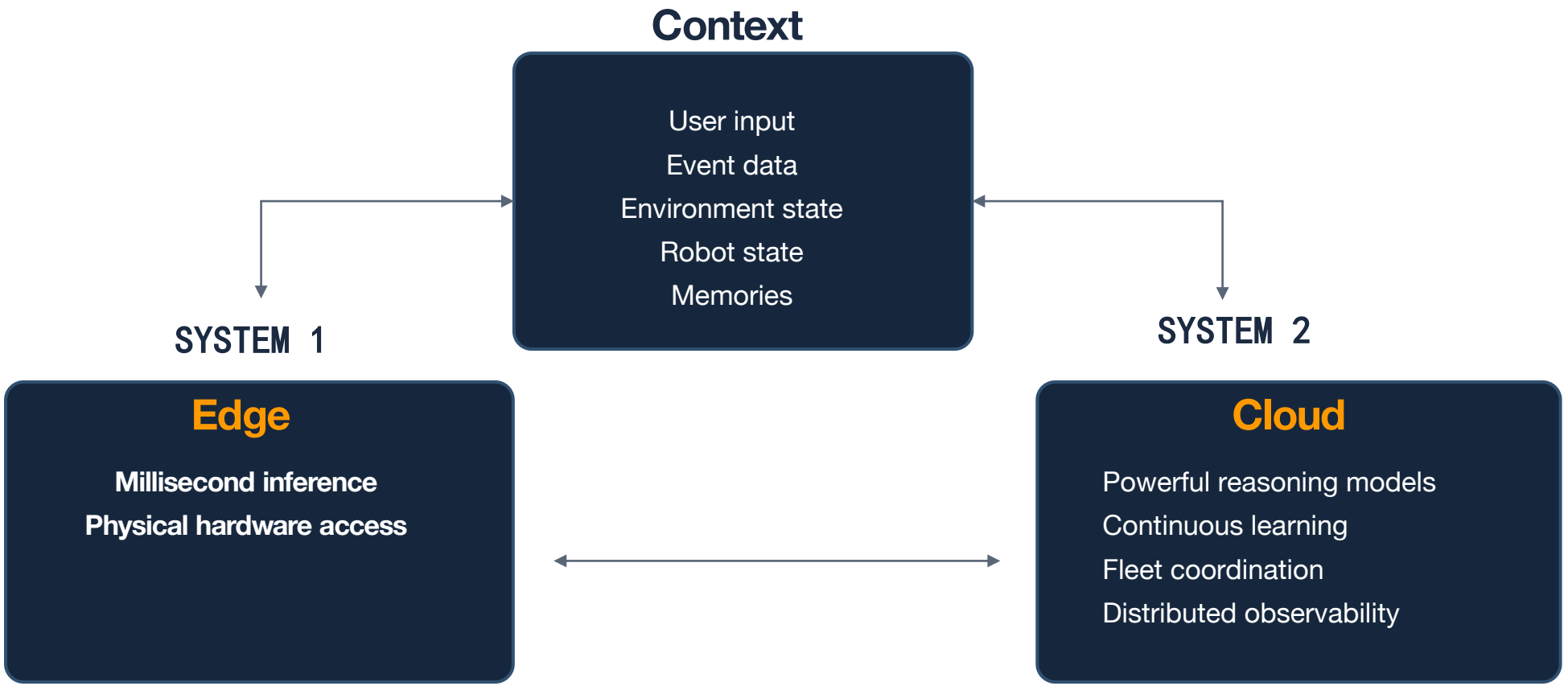


Integration

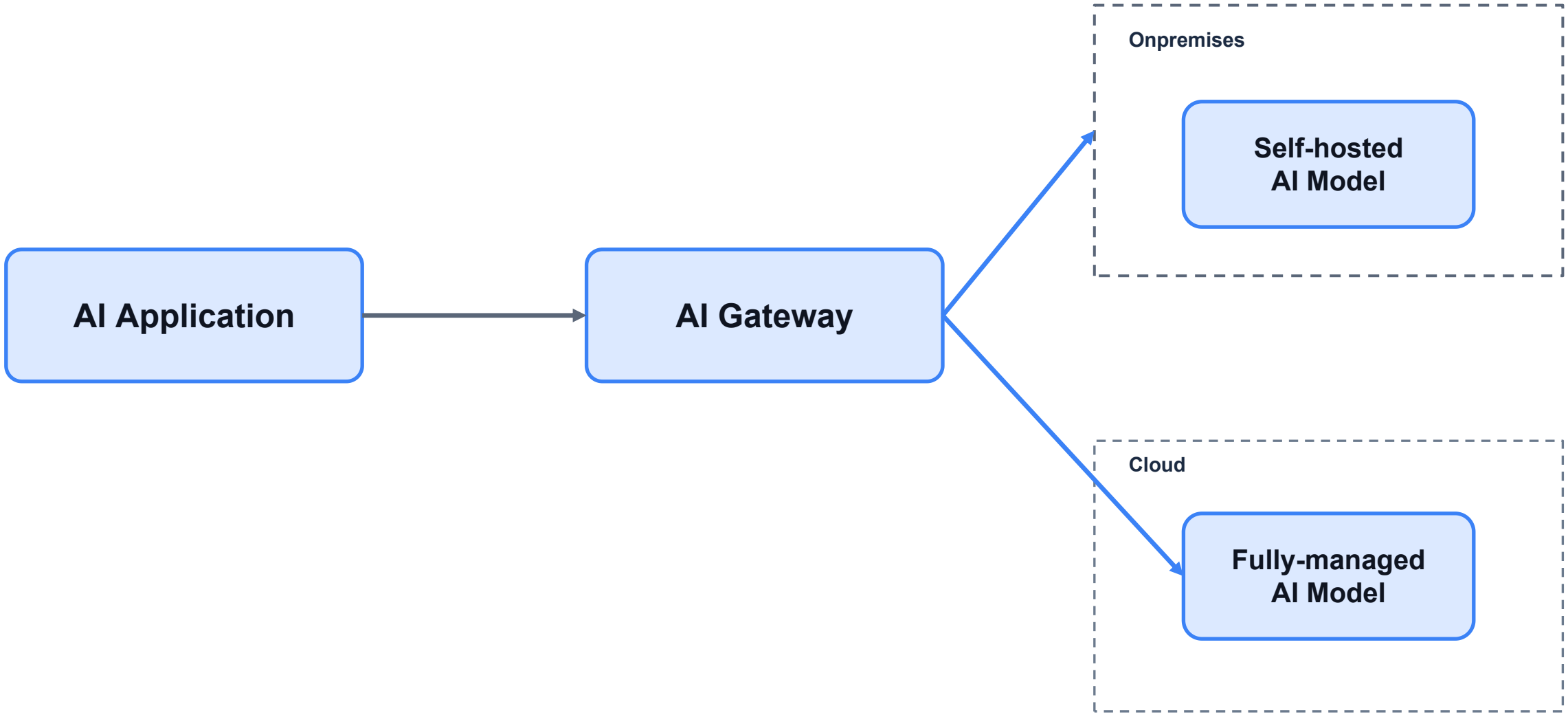


Performance

Dual–system design: fast edge inference + powerful cloud reasoning



Hybrid AI: AI Intelligence



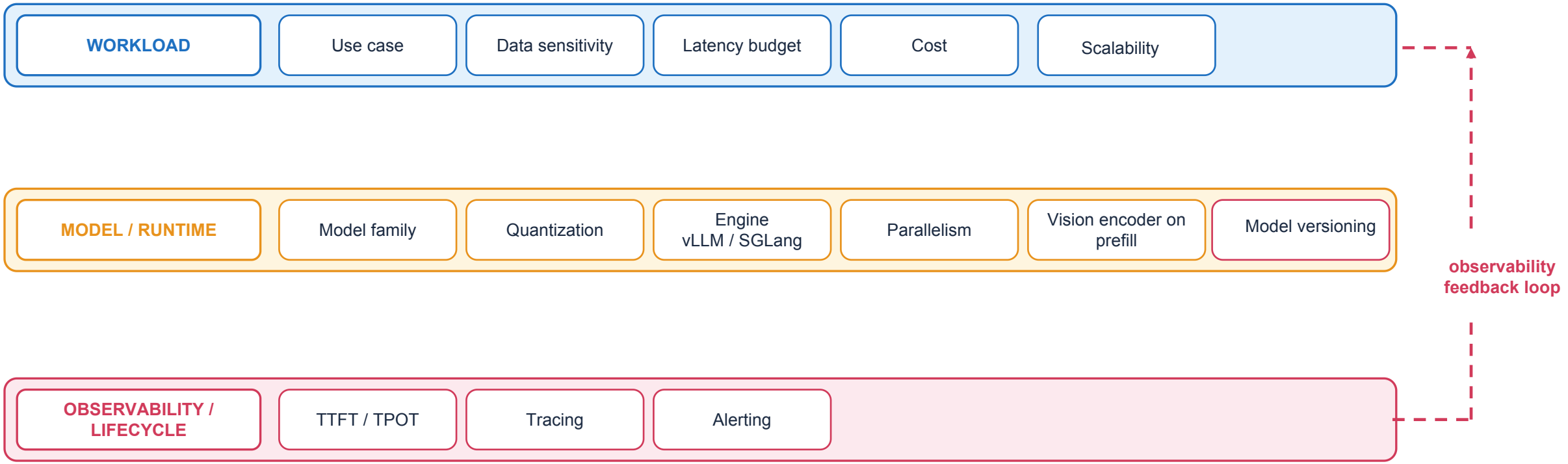
02

Key Architecture Decisions



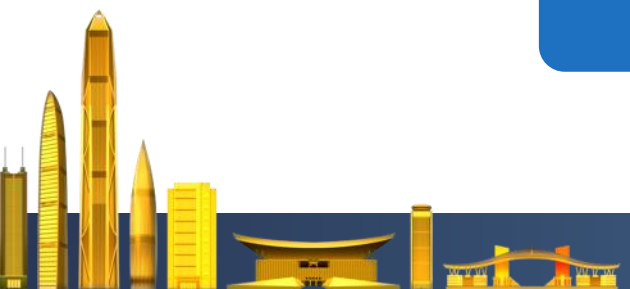
VLM on Kubernetes: Decision Layers

Five layers, decided together. Each layer constrains the next; observability closes the loop.



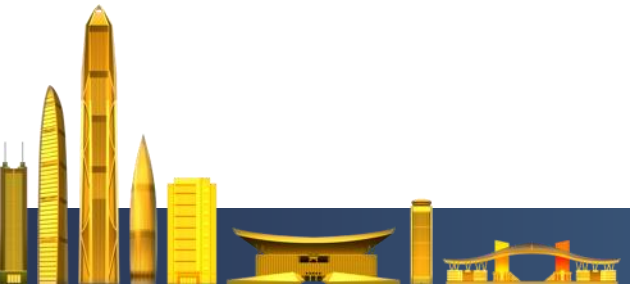
Layer 1: Workload

Use case	Real-time chat tool-assisted agents batch vision analysis, RAG QA
Scalability	How many endpoints to manage?
Latency budget	TTFT target, token rate, p95 / p99 constraints
Input	Text, images, resolution, sample rate



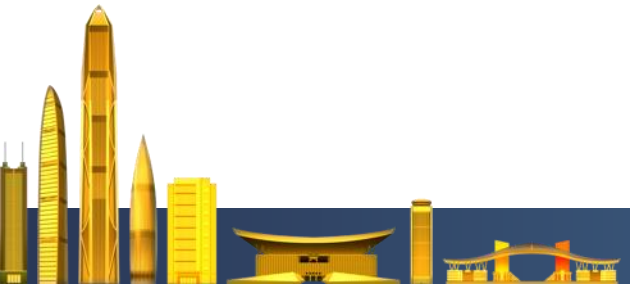
Layer 2: Model & Runtime

Model family	Traditional ML model, LLM, VLM
Quantization	FP16 vs FP8 vs INT8 / INT4
Serving engine	vLLM (default), SGLang, TensorRT-LLM, TGI
Parallelism	Single GPU, tensor parallel, pipeline parallel



Layer 3: Observability & Lifecycle

Metrics	**TTFT, TPOT, tokens/s, queue depth, KV-cache util, cache hit rate** Signals for tuning batch limits and autoscaling
Tracing & logs	**OpenTelemetry, Jaeger, distributed tracing** Debug latency spikes and GPU saturation
Alerting	**GPU saturation, queue build-up, high TTFT, cache-miss spikes** Early detection of capacity or config issues



03

How Edge integrate with Cloud



Amazon EKS portfolio

Cloud

Amazon EKS



AWS FULLY MANAGED

Cloud-connected

Amazon EKS
on Outposts



Amazon EKS
Hybrid Nodes



AWS-MANAGED KUBERNETES
CONTROL PLANE

Cloud-disconnected

Amazon EKS
Anywhere



CUSTOMER-MANAGED
CLUSTER OPERATIONS

Amazon EKS Hybrid Nodes: How it works

NETWORK CONNECTIVITY



AWS Site-to-Site VPN



AWS Direct Connect

INFRASTRUCTURE



Bare-metal servers



Virtual machines

OPERATING SYSTEM

Ubuntu

Bottlerocket

Red Hat Enterprise Linux

Amazon Linux 2023

CREDENTIAL PROVIDER

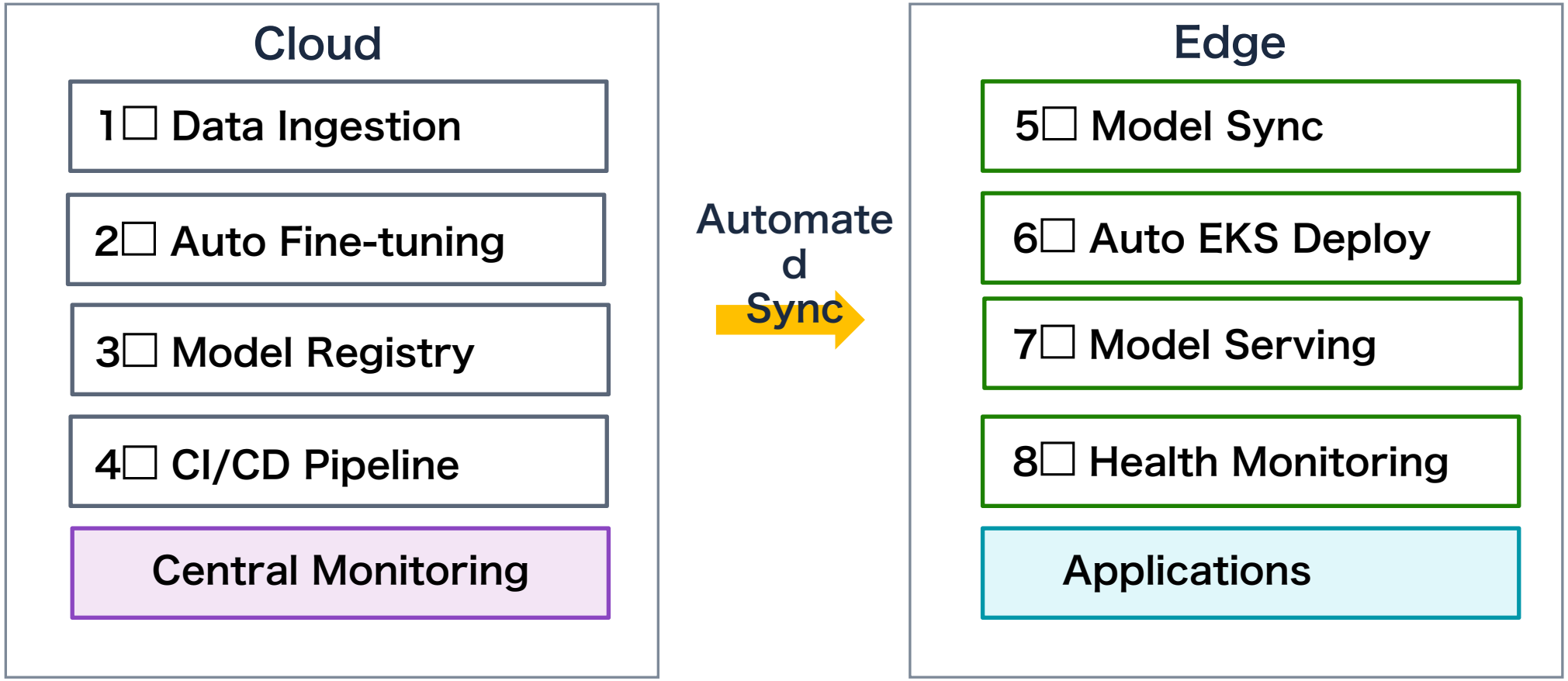


AWS Systems Manager

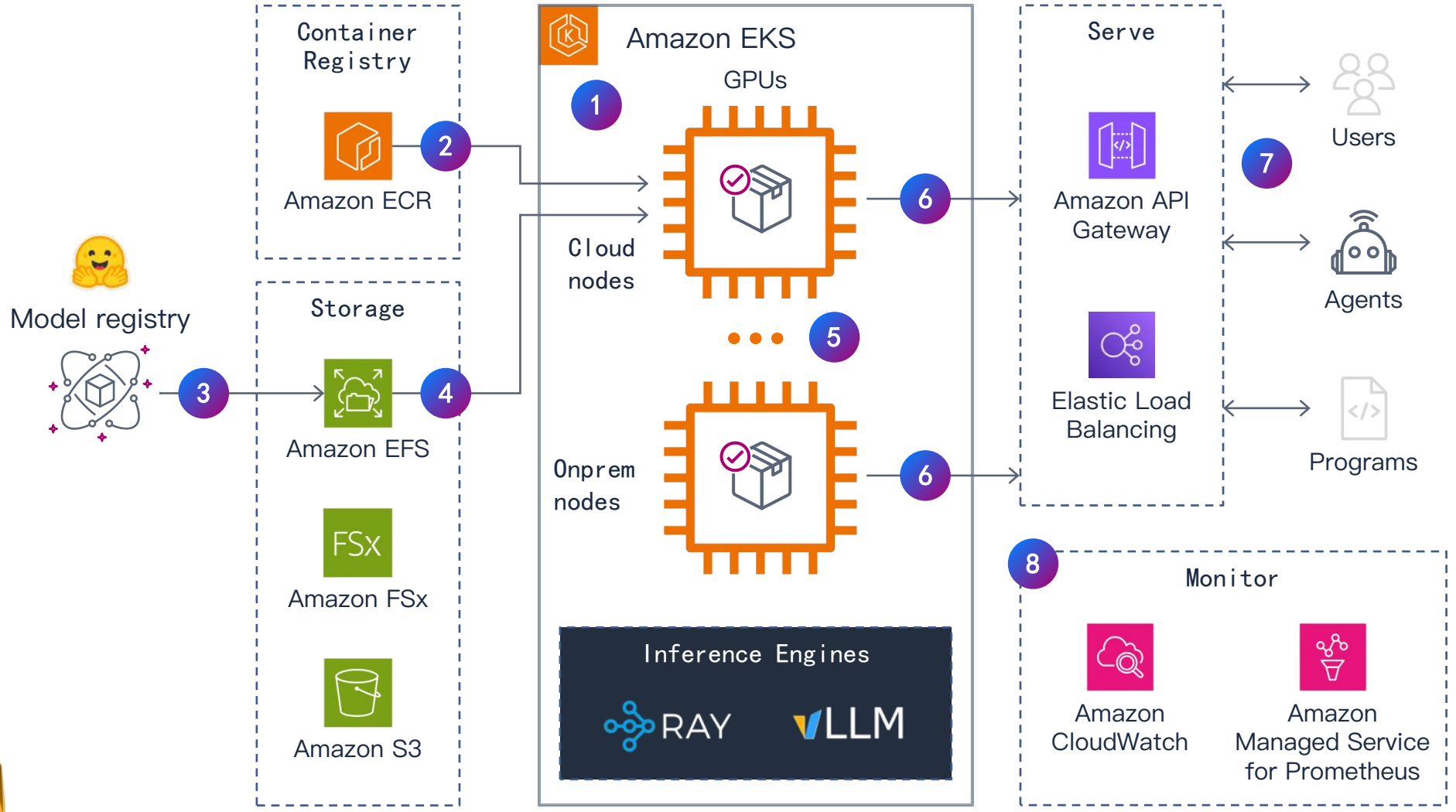


AWS IAM Roles Anywhere

Automated Pipeline



Inference architecture with Hybrid AI



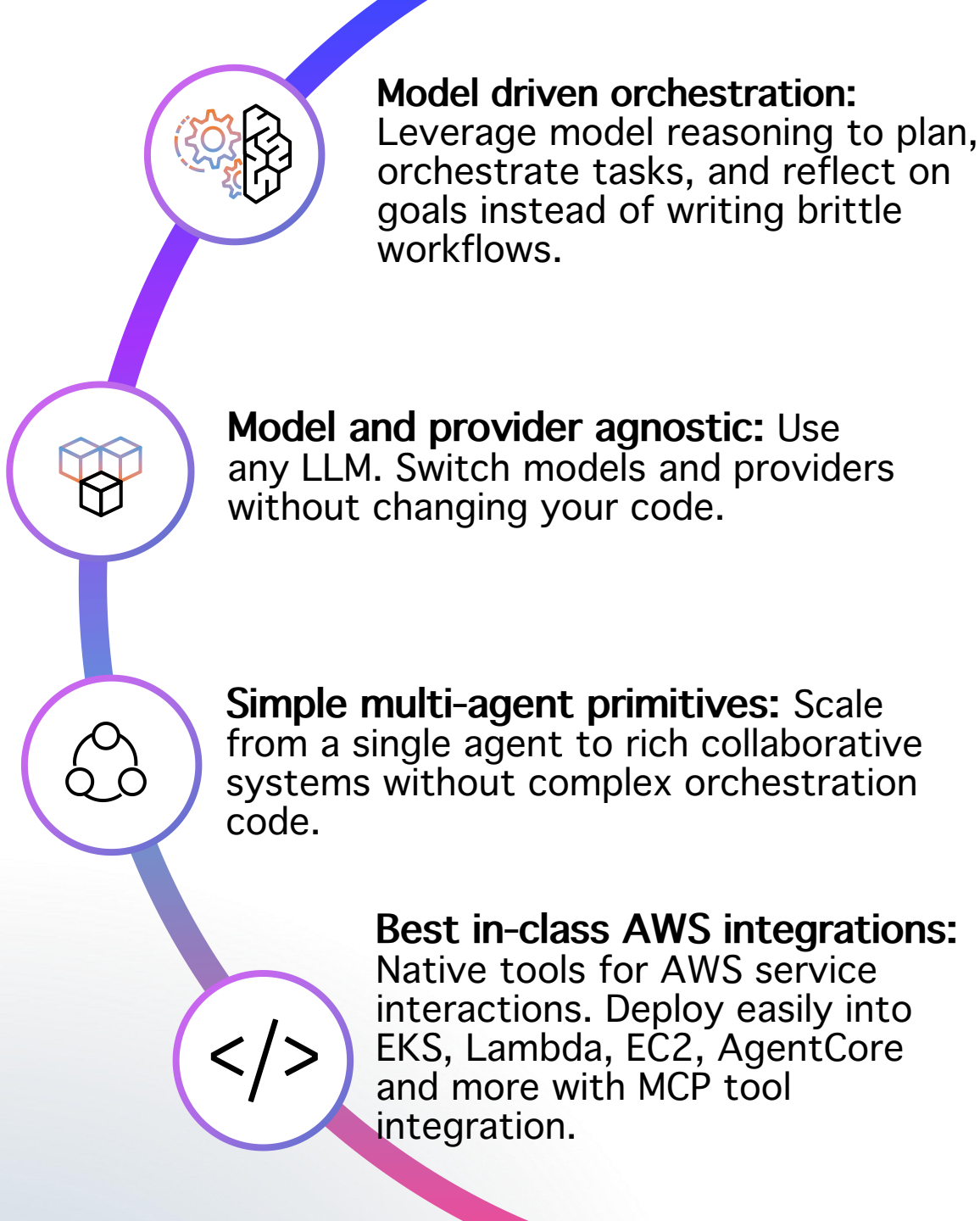
04

Run Agent everywhere



Strands Agents

Strands Agents is an open-source SDK for building agents using just a few lines of code, with first-class support for both Python and TypeScript



Model driven orchestration: Leverage model reasoning to plan, orchestrate tasks, and reflect on goals instead of writing brittle workflows.

Model and provider agnostic: Use any LLM. Switch models and providers without changing your code.

Simple multi-agent primitives: Scale from a single agent to rich collaborative systems without complex orchestration code.

Best in-class AWS integrations: Native tools for AWS service interactions. Deploy easily into EKS, Lambda, EC2, AgentCore and more with MCP tool integration.

STRANDS AGENTS

Deploy strands agents anywhere

Run agents locally
for experimentation

Run agents in a separate environment

Run agents behind an API

Run agents on-prem

Run agents in other clouds

Deploy agents to AWS
for production use

Run agents in any environment where you run Python applications

Real-time safety AI on the floor



AI on EKS

Inference on EKS Workshop

AI/ML on EKS user guide

AI/ML on EKS Terraform Blueprints

Get started with Strands Agents



Docs

Review the user guide, quick start, API documentation for best practices



SDK

Get started building agents and contribute to the open-source project on GitHub



Tools

Access pre-built tools to get started experimenting with agents



Samples

Discover sample agents in our samples repository

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Thanks!



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